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Session : Presentation on Technical Paper

GIS-BASED MAPPING OF RMUS & DTS FOR ASSET RELATIONSHIP MODELING USING AI-BASED MACHINE LEARNING ALGORITHMS IN POWER DISTRIBUTION SYSTEMS

Presented By

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INTRODUCTION

LICENSED UTILITY
NPCL operates as a licensed power distribution utility in Greater Noida

SERVICE AREA
Covering approximately 335 Sq Kilometers in Greater Noida region

CONSUMER BASE
Serving 2 Lakh consumers, predominantly 91% residential

3887 +
MU Annual Energy Sales

750 +
Peak Demands

62 %
Industrial Load Share of Energy Sales

Network Infrastructure

~ 7900 CKM
HT & LT Network Length

78
33/11 kV Substations
~ 958 MVA Capacity

7100+
Distribution Transformers
~ 757 MVA Capacity

Noida Power Company Limited (NPCL) operates as the licensed power distribution utility in Greater Noida since 1993. Utility infrastructure forms the backbone of modern society, with power distribution systems ensuring reliable electricity supply. Managing a large and expanding distribution infrastructure requires accurate asset connectivity mapping and intelligent planning mechanisms.

CONTEXT-WHAT?

Rapid Urbanization

- Greater Noida is a highly emerging area, experiencing rapid urban expansion. Urbanization leads to continuous demographical changes over a minimum span of time , making it necessary for any utility industry to observe these patterns and evaluate their impact on the system performance.

Network Expansion & Infrastructure Growth

- The continuous urban sprawl increases load demand, requiring regular assessment of network capacity and coverage. Hence, it is necessary to monitor the existing electrical infrastructure and plan efficiently for forthcoming requirements.

Inclusion of New Consumers

- Urban transformation has led to continuous introduction of new consumers in society or sectors to the electrical distribution network, leading to increased load demand and higher load concentration across feeders and distribution transformers.

RELEVANCE-WHY IS IT IMPORTANT?

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Efficient Infrastructure Planning and Optimization

- By analyzing load growth and asset performance, electrical infrastructure needs to be planned systematically to ensure optimal utilization of feeders, DTs, RMUs, or substations is done while maintaining system stability.

Proactive Load Forecasting For Societal Development

- Rapid development of new societies has significantly escalated the consumer connections and load demand, thus it is essential to implement structured proactive load forecasting procedure to maintain network reliability & so as to provide best and uninterrupted services to the consumers.

Network Reliability & Back-Feeding Optimization

- The advance planning of RMU-based dual source back-feeding enhances network redundancy. This ensures uninterrupted power supply to be fed to its consumers during feeder outages, maintenance activities or unforeseen events.

AI-Enabled Predictive Planning

- Machine Learning algorithms integrated with the GIS facilitate predictive planning by identifying suitable and optimal locations for installation of electrical assets and infrastructure.

PRESENTATION ON THE TOPIC (1/2)

By integrating AI-ML with GIS data, this project automates the identification of optimal locations for critical electrical assets — RMUs, distribution transformers, and associated infrastructure — enabling strategic, forward-looking network planning across all three connection scenarios.

01

Development of New Society

Planning initiated 6–8 months prior to construction by the facility management so as to ensure feeder capacity, transformer sizing, RMU configuration, and back-feeding before energization.

02

Augmentation of Consumer Load within Existing Society

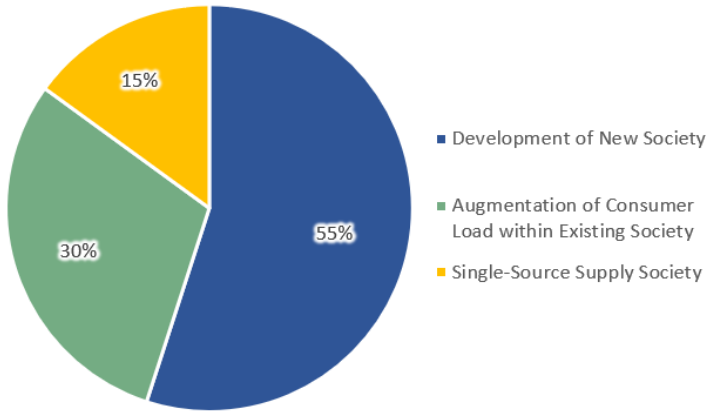
Additional demand evaluated against current feeder & transformer capacity to determine available margin for asset installation.

03

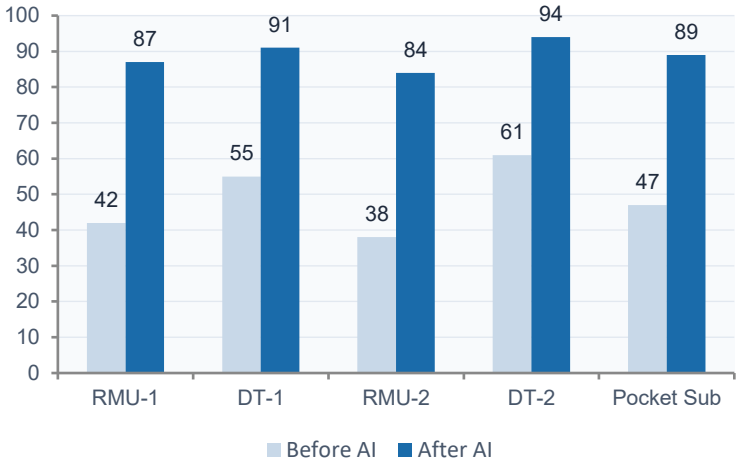
Single-Source Supply Society

Identified and analyzed to determine optimal back-feeding locations, providing a second supply source and improving reliability.

AI-BASED SOCIETAL SCENARIO PLANNING DISTRIBUTION



FEASIBILITY SCORE: BEFORE VS AFTER AI (%)



PRESENTATION ON THE TOPIC (1/2)

DATA COLLECTION	HT/LT Network layout, RMUs, DTs, Load data from existing GIS database
DATA STRUCTURING	Feature classes created with associated attributes
NETWORK TOPOLOGY CREATION	Set Topology Rules, Topology Validation, Built Connectivity Tracing
AOI IDENTIFICATION	Developing new Society, Load Augmentation of consumer within existing one, or Single-Source Society
LOAD FORECASTING	Based on Expected Load Demand & Future growth factor
INTEGRATION OF AI-ML WITH GIS DATA	Creation of Spatial Model to identify optimal location for asset installation automatically
SCENARIO SIMULATION & RELIABILITY TESTING	Simulations created for Testing(Overload, Fault Conditions, etc.)
ANALYSIS	Statistical Analysis to understand feasibility of the scenario
IMPLEMENTATION AT SITE	Implementation of planned scenarios at site

USE CASE / CASE STUDY

NETWORK PLANNING FOR RAJHANS RESIDENCY

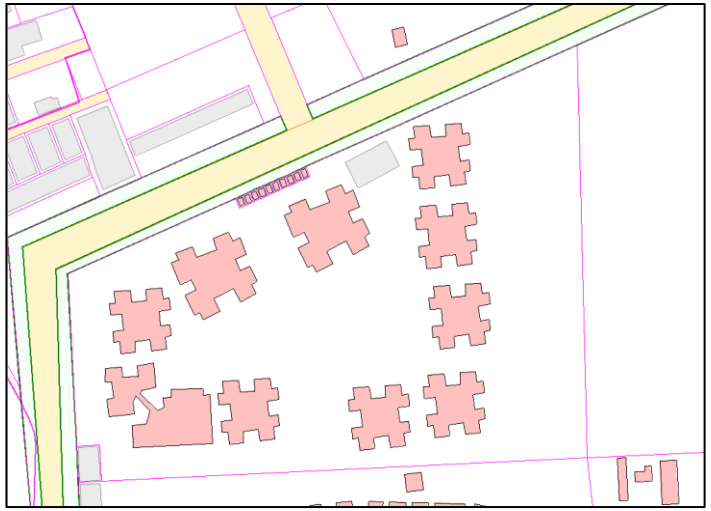
1
- Pre-planning of network infrastructure started prior to 6 months of construction.

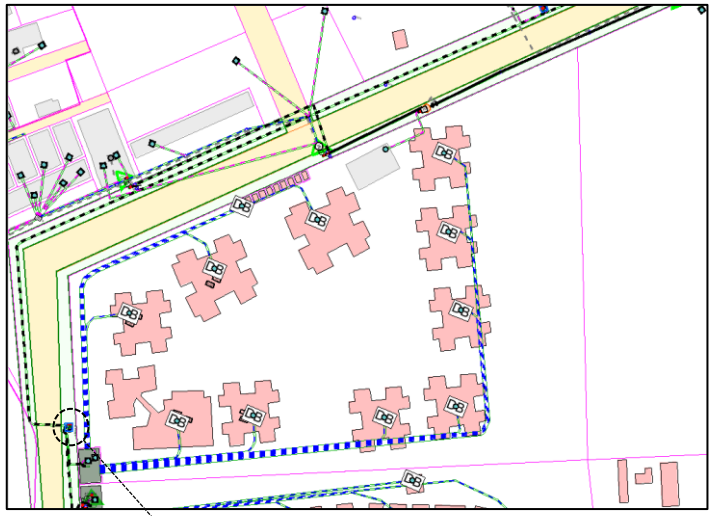
2
- Load-estimation analysis conducted in consideration of count of towers, floors, and anticipated residents.

3
- Automatic identification of Optimal Location for all the network infrastructure done through Machine Learning.

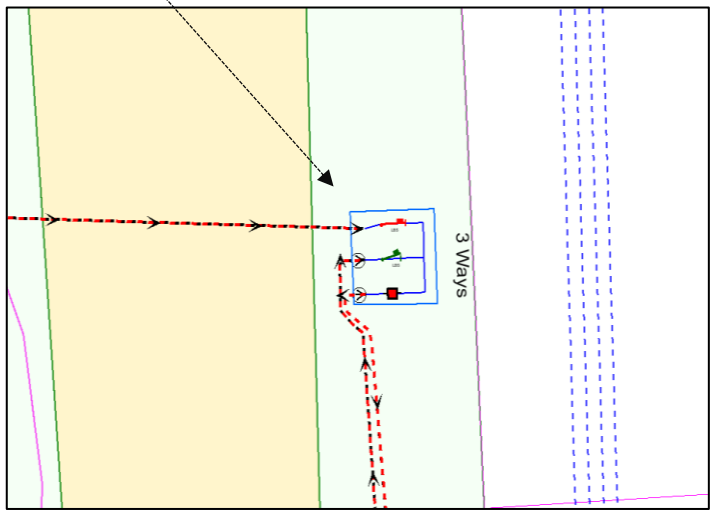
4
- RMU Configuration, Transformer rating, Feeder Connections, & pocket substation installation finalized for the projected load requirements.

5
- Two feeders connected to the RMU to create a dual-source (back-feeding) arrangement, ensuring that if one feeder fails, the other can supply power and maintain uninterrupted electricity.





No. of Towers	No. of Floors	No. of Residentiaries
A	20	80
B	20	80
C	20	80
D	20	80
E	20	80
F	20	80
G	20	80
H	20	120
I	22	88
J	20	80
COMMERCIAL	1 (Ground Floor)	10 Shops



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